VESPAW

Theory and Methods

Variational electronic structure with projector augmented waves

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Version of October 3, 2026

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Chapter 1

Introduction

VESPAW is a plane-wave electronic-structure code based on the projector augmented-wave (PAW) method [1]. It is a fork of CP-PAW, the Car–Parrinello PAW code written by Peter E. Blöchl at Clausthal University of Technology. VESPAW keeps the PAW machinery, the setup generator, the Car–Parrinello dynamics and the input format of CP-PAW, and adds

- a *variational* self-consistent-field (SCF) optimizer, whose energies are upper bounds of the ground-state energy at every iteration (Chapter 6);
- modern iterative eigensolvers (block Davidson, RMM-DIIS, conjugate gradients) next to the original damped dynamics (Chapter 7);
- local hybrid functionals of the PBE0r type and their meta-GGA variants PBE-SL0r and SCAN0r, the SCAN family (SCAN-L and orbital-τ SCAN), and forces and stress for all of them (Chapters 2, 5 and 8);
- quality checks for PAW setups (ghost test, incompleteness warning, kinetic-energy cutoff table), a rebuilt and audited setup library (Chapter 4), and a regression suite (Chapter 9).

License. CP-PAW is distributed under the GNU General Public License, version 3 (GPL-3.0), from its GitHub repository. VESPAW is a derivative of CP-PAW and is therefore distributed under the same license. The older notice "(C) Copyright Clausthal University of Technology, licensed materials" still found in the headers of the inherited source files predates the GPL release.

How to read this book. Chapters 2–4 summarize the physics that VESPAW implements: density-functional theory, plane waves and the PAW method. Chapter 5 treats the local hybrid functionals. Chapters 6 and 7 describe how the Kohn–Sham equations are solved; this is where VESPAW differs most from other codes. Chapter 8 covers forces, stress and molecular dynamics, and Chapter 9 the structure of the program. Input keywords are given in `typewriter` font; their complete description is in the input manual `doc/manual.tex`. Hartree atomic units are used throughout unless stated otherwise; cutoff energies are quoted in Rydberg, as in the input.

Pitfalls. Many sections end with a short list of pitfalls. They record errors that were actually found in the code and fixed, so that the reader understands which quantities are delicate and how they are tested.

Chapter 2

Density-functional theory

2.1 Kohn–Sham equations

By the Hohenberg–Kohn theorem [2] the ground-state energy of N interacting electrons in an external potential v_{ext} is a functional of the density $n(\boldsymbol{r})$. Kohn and Sham [3] write it in terms of non-interacting orbitals ψ_n with occupations f_n,

$$E[n] = \sum_n f_n \langle \psi_n | -\tfrac{1}{2}\nabla^2 | \psi_n \rangle + \int d^3r\, v_{\mathrm{ext}} n + \frac{1}{2} \iint d^3r\, d^3r' \, \frac{n(\boldsymbol{r}) n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} + E_{\mathrm{xc}}[n], \qquad n(\boldsymbol{r}) = \sum_n f_n |\psi_n(\boldsymbol{r})|^2. \tag{2.1}$$

Minimizing with orthonormal orbitals gives the Kohn–Sham equations

$$\left[-\tfrac{1}{2}\nabla^2 + v_{\mathrm{eff}}(\boldsymbol{r}) \right] \psi_n = \epsilon_n \psi_n, \qquad v_{\mathrm{eff}} = v_{\mathrm{ext}} + v_{\mathrm{H}}[n] + \frac{\delta E_{\mathrm{xc}}}{\delta n}. \tag{2.2}$$

Because v_{eff} depends on the density, the equations must be solved self-consistently (Chapter 6). A self-consistent solution is a stationary point of E only if the potential is the functional derivative of the energy. This condition is easy to state and easy to violate in an implementation; VESPAW tests it routinely (Sec. 6.6).

2.2 Exchange-correlation functionals

LDA and GGA. The local-density approximation uses the exchange-correlation energy per electron of the homogeneous electron gas, $E_{\mathrm{xc}} = \int n\, \epsilon_{\mathrm{xc}}(n)$ (PZ [4], PW92 [5]). Generalized gradient approximations add a dependence on $|\nabla n|$; the default in VESPAW is PBE [6]. All functionals are selected by `!CONTROL!DFT TYPE`, as a number or a name, e.g. `TYPE='PBE'`; the VASP `GGA` tags are accepted as names. Table 2.1 lists them.

Parameters. The constants follow the original publications and agree with libxc and VASP: $\beta = 0.06672455060314922$ in PBE correlation and in $\mu = \beta\pi^2/3$ of PBE exchange (before October 2026 VESPAW used $\beta = \nu C_c(0) = 0.0667263$, kept by `TYPE=-10` together with the pre-2024 reduced gradient, t^2 too small by a factor 0.62); PW91 keeps $\beta = \nu C_c(0)$, as in VASP; PBEsol uses $\mu = 10/81$ and $\beta = 0.046$; SCAN-L uses the optimized Perdew–Constantin parameters $a = 1.784720$, $b = 0.258304$ of Ref. [16]; the spin interpolation of SCAN correlation uses $G_c(\zeta) = [1 - 2.363(d_x(\zeta) - 1)](1 - \zeta^{12})$, with 2.363 as in VASP and libxc (the SCAN supplement quotes 2.3631). Unit conversions use CODATA 2018 (1 Hartree = 27.211386 eV, 1 bohr = 0.529177211 Å). Plane-wave hybrids with full exact exchange (PBE0, HSE06, SCAN0, B3LYP) are not yet available.

Table 2.1: Exchange-correlation functionals of VESPAW. "libxc" marks functionals whose energy density agrees with libxc 7.1.2 (and the VASP implementation) to $\sim 10^{-15}$ at every point of a grid in density, gradient, spin polarization and α (October 2026). The local hybrids are described in Chapter 5.

TYPE	name	functional	VASP	libxc
LDA				
1	PZ	Perdew–Zunger [4] of Ceperley–Alder [7]		✓
2	PW92	Perdew–Wang 92 [5]		✓
3		Slater exchange only		✓
4		X_α, $\alpha = 0.7$		
GGA				
10	PBE, PE	PBE [6]	GGA=PE	✓
11	RPBE, RP	RPBE [8]	GGA=RP	✓
12	PW91, 91	PW91 exchange and correlation [9]	GGA=91	✓
13	REVPBE, RE	revPBE, $\kappa = 1.245$ [10]	GGA=RE	✓
14	PBESOL, PS	PBEsol [11]	GGA=PS	✓
5001	BLYP	B88 [12] + LYP [13]		✓
81		BP86: PZ + B88 + P86 [14]		✓
8		B88 + P86 on PW92 local correlation (non-standard)		
7, 71, 9		B88 exchange with PW92, PZ or PBE correlation		✓
6		B88 exchange only		✓
meta-GGA				
21	SCAN	SCAN, orbital τ [15] (with TSCANDELOC=T)	METAGGA=SCAN	✓
20	SCANL	SCAN-L, deorbitalized [16, 17]	METAGGA=SCANL	✓
local hybrids (Chapter 5)				
	PBE0r	PBE + onsite exact exchange		
	PBE-SL0r	PBE; onsite SCAN-L/exact exchange		
	SCAN0r	SCAN + onsite exact exchange		
for testing and reproducing old results				
−10		PBE before the 2024/2026 corrections (see text)		
−12		PBE exchange + PW91 correlation (old type 12)		
5002		PBE with exchange scaled by 0.75		
10001–7		single components (PZ, PW92, P86, PBE, PW91 correlation; PBE, RPBE exchange)		

Meta-GGAs: SCAN, SCAN-L and TSCAN. The SCAN functional [15] depends in addition on the kinetic energy density $\tau = \frac{1}{2}\sum_n f_n |\nabla\psi_n|^2$ through the iso-orbital indicator

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \qquad \tau_W = \frac{|\nabla n|^2}{8n}, \qquad \tau_{\text{unif}} = \tfrac{3}{10}(3\pi^2)^{2/3} n^{5/3}. \tag{2.3}$$

VESPAW provides two routes:

- **SCAN-L** (TYPE='SCANL'), the deorbitalized SCAN of Mejía-Rodríguez and Trickey [16, 18], in which α is approximated from n, $|\nabla n|$ and $\nabla^2 n$. The functional depends on the Laplacian of the density, so its potential contains second-derivative terms (see below).
- **TSCAN** (TYPE='SCAN' with TSCANDELOC=T), the orbital-dependent SCAN with the true τ. The Hamiltonian then contains the operator $-\frac{1}{2}\nabla \cdot v_\tau \nabla$ with $v_\tau = \partial e_{\text{xc}}/\partial\tau$. In the PAW spheres the one-center τ is built from the partial waves, including its non-spherical parts.

Functional derivatives. For a semilocal energy $E = \int e(n, |\nabla n|^2, \nabla^2 n)\, d^3r$ the potential is the full Euler–Lagrange derivative

$$v = \frac{\partial e}{\partial n} - \nabla \cdot \Big(2 \frac{\partial e}{\partial |\nabla n|^2} \nabla n \Big) + \nabla^2 \frac{\partial e}{\partial \nabla^2 n}. \tag{2.4}$$

For GGAs the last term is absent; for SCAN-L it is essential. Where the energy carries a spatial weight (the double counting of the local hybrids, Sec. 5.4), the weight stands *inside* the derivatives. In the spherical one-center expansion the divergence becomes $r^{-2}(r^2 A)'$; in the non-spherical case the weak form is used, projecting $\partial e/\partial \nabla n$ onto the radial and tangential gradients of each spherical harmonic.

Pitfall. Before September 2026, the one-center SCAN-L and TSCAN potentials kept only the first term of Eq. (2.4). Self-consistent solutions were then not stationary points of the energy: the variational optimizer found lower energies without reaching self-consistency, and for TSCAN it did not converge at all. The consistency test of Sec. 6.6 exposed both.

2.3 Spin

Collinear spin polarization (`!STRUCTURE!OCCUPATIONS NSPIN=2`) uses two sets of orbitals and the spin densities $n_\uparrow, n_\downarrow$; the code works internally with total and spin density (n, m). Non-collinear magnetism uses two-component spinors. For exchange, the spin-scaling relation

$$E_x[n_\uparrow, n_\downarrow] = \tfrac{1}{2}\big(E_x[2n_\uparrow] + E_x[2n_\downarrow]\big) \tag{2.5}$$

is exact and allows any spin-unpolarized exchange functional to be applied to a spin-polarized density. VESPAW uses it for the exchange of the SCAN family, on the grid and in the one-center terms. Correlation does not obey Eq. (2.5); SCAN correlation is evaluated for the spin densities with the spin-interpolation of Ref. [15]: $\phi(\zeta)$ in the gradient term, $G_c(\zeta)$ in the single-orbital limit, and α divided by $d_s(\zeta) = [(1+\zeta)^{5/3} + (1-\zeta)^{5/3}]/2$ with the total τ. For SCAN-L, the spin-resolved kinetic energy densities are deorbitalized separately, $\tau[n_\uparrow, n_\downarrow] = \frac{1}{2}(\tau[2n_\uparrow] + \tau[2n_\downarrow])$, as in libxc.

2.4 Finite temperature

For metals and for systems with near-degenerate states, VESPAW minimizes the Mermin free energy [19]

$$A = E[\{\psi_n\}, \{f_n\}] - TS[\{f_n\}], \qquad S = -k_B \sum_n \big[f_n \ln f_n + (1 - f_n) \ln(1 - f_n) \big], \tag{2.6}$$

with Fermi occupations $f_n = [1 + e^{(\epsilon_n - \mu)/k_B T}]^{-1}$ at fixed electron number. The electronic temperature is set in `!CONTROL!MERMIN T[K]`. It acts as a smearing parameter; the free energy A, not E, is the variational quantity.

Chapter 3

Plane waves and Brillouin-zone sampling

3.1 Bloch states and plane waves

In a crystal with lattice vectors $\boldsymbol{T}$ the orbitals are Bloch states, $\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{T}) = e^{i\boldsymbol{k}\cdot\boldsymbol{T}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$, which are expanded in plane waves,

$$\tilde{\psi}_{n\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, e^{i(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \qquad \tfrac{1}{2}|\boldsymbol{k}+\boldsymbol{G}|^2 \le E_{\mathrm{cut}}. \tag{3.1}$$

In PAW only the smooth pseudo wave functions $\tilde{\psi}$ are expanded in plane waves (Chapter 4).

3.2 Cutoffs and FFT grids

The wave-function cutoff is `!CONTROL!FOURIER EPWPSI` in Rydberg. The density contains products of wave functions and would need four times the cutoff; VESPAW uses `EPWRHO = CDUAL`×`EPWPSI`. `CDUAL=4` is exact, `CDUAL=2` is usually sufficient and cheaper. Density and local potential live on a real-space grid; the Hamiltonian is applied to a wave function by transforming it to real space with a fast Fourier transform, multiplying by the potential and transforming back. In VESPAW these transforms are batched over blocks of 32 bands.

Energies converge from above with the cutoff, but energy *differences*, forces and stress converge faster. The required cutoff is set by the hardest PAW setup in the system; the setup generator prints an estimate (Sec. 4.4).

3.3 k-point sampling

Brillouin-zone integrals are replaced by sums over a Monkhorst–Pack mesh [20], `!STRUCTURE!KPOINTS DIV=`n_1 n_2 n_3, with an optional shift, or by a target density `R`. Symmetry and time reversal ($\psi_{-\boldsymbol{k}} = \psi^*_{\boldsymbol{k}}$) reduce the number of irreducible points. Metals need dense meshes together with a finite electronic temperature; insulators converge quickly.

3.4 The Γ point

At $\boldsymbol{k} = 0$ the wave functions can be chosen real. CP-PAW exploits this by packing two real bands into one complex array ("super wave functions"), which halves the number of FFTs. An alternative half-sphere storage (`GAMMASUPER`) is experimental and not yet correct.

Chapter 4

The projector augmented-wave method

4.1 The PAW transformation

The PAW method [1] maps the true all-electron (AE) valence orbitals ψ_n, which oscillate strongly near the nuclei, onto smooth pseudo (PS) orbitals $\tilde{\psi}_n$ by a linear transformation

$$|\psi_n\rangle = |\tilde{\psi}_n\rangle + \sum_{R,i} \big(|\phi_{R,i}\rangle - |\tilde{\phi}_{R,i}\rangle\big)\langle \tilde{p}_{R,i}|\tilde{\psi}_n\rangle. \tag{4.1}$$

Here $\phi_{R,i}$ are AE partial waves (solutions of the atomic Schrödinger equation at reference energies), $\tilde{\phi}_{R,i}$ the corresponding PS partial waves, which agree with $\phi_{R,i}$ outside an augmentation radius r_c, and $\tilde{p}_{R,i}$ projector functions, localized within r_c and dual to the PS partial waves, $\langle \tilde{p}_{R,i}|\tilde{\phi}_{R,j}\rangle = \delta_{ij}$. The index i collects the angular momentum (ℓ, m) and a counter ν over partial waves of the same ℓ. The number of partial waves per ℓ is set by `!STRUCTURE!SPECIES NPRO`.

Expectation values split into a plane-wave part and one-center corrections,

$$\langle A\rangle = \sum_n f_n \langle\tilde{\psi}_n|\tilde{A}|\tilde{\psi}_n\rangle + \sum_R \sum_{ij} D_{R,ij}\big(\langle\phi_{R,j}|A|\phi_{R,i}\rangle - \langle\tilde{\phi}_{R,j}|\tilde{A}|\tilde{\phi}_{R,i}\rangle\big), \qquad D_{R,ij} = \sum_n f_n \langle\tilde{p}_{R,i}|\tilde{\psi}_n\rangle\langle\tilde{\psi}_n|\tilde{p}_{R,j}\rangle, \tag{4.2}$$

and the total energy becomes $E = \tilde{E} + \sum_R (E^1_R - \tilde{E}^1_R)$: a plane-wave term plus, for each atom, the difference of an AE and a PS one-center energy evaluated on radial grids. The orbitals are orthonormal with respect to the overlap operator

$$\hat{O} = 1 + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta O_{R,ij} \langle\tilde{p}_{R,j}|, \qquad \Delta O_{R,ij} = \langle\phi_{R,i}|\phi_{R,j}\rangle - \langle\tilde{\phi}_{R,i}|\tilde{\phi}_{R,j}\rangle, \tag{4.3}$$

so the Kohn–Sham equations take the generalized form $\tilde{H}|\tilde{\psi}_n\rangle = \epsilon_n \hat{O}|\tilde{\psi}_n\rangle$ with

$$\tilde{H} = -\tfrac{1}{2}\nabla^2 + \tilde{v}(\boldsymbol{r}) + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta H_{R,ij} \langle\tilde{p}_{R,j}|. \tag{4.4}$$

4.2 Compensation charges, core and local potential

The electrostatics of the one-center densities is handled by compensation charges $\hat{n}_R$ (Gaussians of width set by `RCSM/RCOV`) that reproduce the multipole moments of $n^1_R - \tilde{n}^1_R$, so that the long-range electrostatics is carried by the smooth density $\tilde{n} + \hat{n}$ on the plane-wave grid. The one-center expansions are truncated at $\ell \le$ `LRHOX`. Core electrons are frozen in their atomic state. For the exchange-correlation energy, which is non-linear in the density, a pseudized core density (`!CORE POW RC`) is added to the smooth density. The local pseudopotential is constructed by pseudizing the AE potential inside a radius (`!POT POW RC`).

4.3 Setup construction

A PAW *setup* contains, for one element, the partial waves, projectors, core densities and local potential. VESPAW builds setups at run time from a short parameter block (`!SPECIES!AUGMENT`), or reads them from the library `vespaw_pot`. Three construction types exist:

- `HBS`: the scheme of Blöchl's later CP-PAW versions. The first partial wave of each ℓ is the lowest valence state; further partial waves are obtained from the nodeless ("$\dot{\phi}$-like") construction at higher energies, bounded by a box of radius `RBOX/RCOV`.
- `NDLSS`: an older nodeless scheme.
- `KERKER`: Kerker-type pseudization.

The radii are given in units of the covalent radius of the element: `RCL/RCOV` for the partial waves, `RCSM/RCOV` for the compensation charge, `RC/RCOV` in `!POT` and `!CORE`.

Semicore states. When shallow core states are treated as valence (e.g. Cu $3s3p$, `ZV=19`), an angular momentum has two valence states, $3s$ and $4s$. Each valence bound state then receives its own partial wave at its own energy; only further partial waves are box states. With a semicore state below it, the PS $4s$ orbital need not have a node, because it is orthogonal to the PS $3s$ only through $\hat{O}$. The setup test therefore identifies PS states by their energy, not by their node count.

4.4 Quality checks

A setup is an approximation, and a bad one fails silently: calculations converge to wrong answers. The generator therefore reports three checks.

Ghost test. The atomic PAW Hamiltonian of each ℓ is diagonalized in a basis of 120 spherical Bessel functions inside a large sphere. Its lowest levels must match the AE valence levels. A level far below the lowest AE reference is a *ghost state*; the generator prints `WARNING: GHOST STATE IN SETUP`. The overlap matrix in this basis must be positive definite; if it is not, the setup is unusable.

Incompleteness warning. With only one partial wave per ℓ, the local potential may bind a state far below the valence level whose wave function lies largely outside the span of the projectors. Such components receive no one-center correction. In a solid they can sink into the potential well of the compensation charge, and the calculation converges to a collapsed state far below the physical minimum. This happened for NiO with a single-projector Ni setup: the collapse deepened with the cutoff and disappeared with a second s projector. The generator prints `WARNING: INCOMPLETE SETUP` when the lowest state of the local potential lies more than 0.5 H below the valence reference with a single projector.

Kinetic-energy cutoff table. As in the VASP setup generator, the kinetic energy of each PS valence orbital that lies above a cutoff is computed from its Fourier transform, weighted by the occupation and summed. The generator prints the error as a function of the cutoff, from 20 eV to about 290 Ry, and suggests `ENMAX` and `ENMIN` (errors of 6 and 13 meV), or states that they are not reached. The PS orbitals are tapered to zero before the box wall to avoid a spurious kink.

4.5 The setup library

`vespaw_pot` contains one species block per element and variant. Its audit (`vespaw_pot/AUDIT.md`) lists each setup with its status (OK, incomplete, ghost, unusable) and the recommended replacement. The flagged setups were rebuilt as HBS setups with two projectors per ℓ. Validation in solids shows that the frozen-semicore $3d$ setups (Fe, Cu) reproduce AE-PBE lattice constants only to about 1%, and that the error grows with the augmentation radius; a semicore Cu setup with small radii reaches 0.5% at a high cutoff. Accurate soft $3d$ setups are an open problem (see the audit).

Chapter 5

Local hybrid functionals

Hybrid functionals admix a fraction a_x of exact exchange and cure much of the band-gap problem of semilocal functionals, but in a plane-wave basis exact exchange is one to two orders of magnitude more expensive than a GGA. Local hybrids of the PBE0r type [21, 22] keep only the onsite part of the exchange correction, evaluated in a basis of local orbitals, at nearly semilocal cost.

5.1 Local orbitals

Each atom carries natural tight-binding orbitals (NTBOs) χ_a [21, 23]. Inside the augmentation sphere of radius `RAUG/RCOV` the head function of an orbital is an augmented screened Hankel function, a combination of an AE partial wave ϕ and its energy derivative $\dot\phi$,

$$K^{\mathrm{aug}} = \phi\, k_\phi + \dot\phi\, k_{\dot\phi}, \tag{5.1}$$

matched to a Hankel function with κ^2 set by `K2`. In the spheres of the neighbours it is continued by screened Bessel functions $\bar J^{\mathrm{aug}} = \dot\phi\, j_{\dot\phi}$ with the structure constants $\bar S$. Exponential tails (`TAILLAMBDA`, `RTAIL/RCOV`) and a cutoff (`RTAILCUT/RCOV`) make the orbitals short ranged. The orbitals considered per ℓ are selected by `NOFL`.

5.2 Projection onto the local orbitals

The onsite density matrix $D^\sigma_{ab} = \sum_n f_{n\sigma}\langle\chi_a|\psi_{n\sigma}\rangle\langle\psi_{n\sigma}|\chi_b\rangle$ requires the coefficients of the wave functions in the local-orbital basis. Inside the spheres the one-center expansion $\psi^1 = \sum_i \phi_i\langle\tilde p_i|\tilde\psi\rangle$ is fitted by local orbitals in the least-squares sense,

$$\boldsymbol{c} = \big[\langle\chi|\Theta|\chi\rangle\big]^{-1}\langle\chi|\Theta|\phi\rangle\langle\tilde p|\tilde\psi\rangle, \tag{5.2}$$

where Θ restricts integrals to the spheres. Both overlaps are computed with the exact χ inside the spheres, i.e. with K^{aug} and $\bar J^{\mathrm{aug}}$ as radial functions.

Pitfall. The original implementation represented χ by its partial-wave expansion $\langle\tilde p|\tilde\chi\rangle$. The $\dot\phi$ component of χ converges slowly in that expansion, so the fitted orbital lacked norm and the density matrix overcounted charge: 13–15 electrons on Ni in NiO instead of about 7. The hybrid energy and the gap then changed with the number of projectors (NiO PBE0r gap 3.96, 4.08, 4.10 eV for 2, 3, 4 d projectors). With the exact overlaps they are converged (2.99, 3.00, 3.00 eV). The old fit can be restored with the environment variable `VESPAW_NTBO_OLDPROJ=1` for comparison.

5.3 PBE0r

PBE0r keeps only the onsite terms of the PBE0 exchange correction,

$$E_{\rm xc}^{PBE0r} = E_{\rm xc}^{\rm PBE}[n] + a_x\big(E_x^{\rm HF,1c} - E_x^{\rm PBE,1c}\big), \qquad E_x^{\rm HF,1c} = -\frac{1}{2}\sum_\sigma \sum_{abcd} D^\sigma_{ac} D^\sigma_{bd} U_{abcd}, \tag{5.3}$$

with the onsite Coulomb tensor U of the local orbitals. The double-counting term $E_x^{\rm PBE,1c}$ is the PBE exchange of the local-orbital density, obtained from the full one-center density with the weight $w(r) = (n_{\rm loc}/n_{\rm all})^2$. Core–valence exchange can be included (`CV=T`). The mixing a_x is set globally by `!CONTROL!DFT!NTBO HFWEIGHT` and per species by `LHFWEIGHT`. The exchange correction enters the Hamiltonian in two places: the one-center PAW Hamiltonian ΔH (derivative with respect to the full one-center density) and an operator in the local-orbital basis (derivative with respect to D).

5.4 PBE-SL0r

PBE-SL0r (called SCAN0r before October 2026) replaces the semilocal onsite exchange by SCAN-L exchange while keeping PBE for the delocalized part and for all correlation:

$$\begin{aligned} E_{\rm xc}^{PBE-SL0r} &= E_{\rm xc}^{\rm PBE}[n] - E_x^{\rm PBE,1c} + a_x E_x^{\rm HF,1c} + (1-a_x) E_x^{\rm SCAN,1c} \\ &= E_{\rm xc}^{\rm PBE}[n] + \underbrace{\big(E_x^{\rm SCAN,1c} - E_x^{\rm PBE,1c}\big)}_{\text{swap term}} + a_x\big(E_x^{\rm HF,1c} - E_x^{\rm SCAN,1c}\big). \end{aligned} \tag{5.4}$$

The swap term is not scaled by a_x. PBE-SL0r is selected by `!NTBO MODUS='HYBRID' HFWEIGHT=`a_x `DCTYPE='SCANL'` with a PBE base functional. A consistent implementation requires:

1. the swap term in both Hamiltonian channels (one-center and local-orbital), at full weight;
2. the full Euler–Lagrange potential of the onsite SCAN-L exchange, Eq. (2.4), with the weight w inside the derivatives;
3. exact spin scaling, Eq. (2.5). Evaluating the onsite SCAN-L exchange for the spin-summed density replaces spin-polarized PBE exchange by unpolarized SCAN-L exchange in the swap term and removes a fraction $1 - a_x$ of the onsite exchange splitting; magnetic gaps collapse (NiO 1.0 instead of 2.0 eV);
4. the exact local-orbital projection of Sec. 5.2.

All four were found by checking that the variational optimizer and damped dynamics reach the same state (Sec. 6.6).

5.5 SCAN0r

SCAN0r uses SCAN with the orbital kinetic energy density everywhere, for the plane-wave density and in the spheres, for exchange and correlation, and replaces a fraction a_x of the onsite SCAN exchange of the local orbitals by exact exchange:

$$E_{\rm xc}^{SCAN0r} = E_{\rm xc}^{\rm SCAN}[n,\tau] + a_x\big(E_x^{\rm HF,1c} - E_x^{\rm SCAN,1c}\big). \tag{5.5}$$

It is the meta-GGA analogue of PBE0r, Eq. (5.3), without a swap term, and is selected by `TYPE='SCAN' TSCANDELOC=T` with `!NTBO MODUS='HYBRID' HFWEIGHT=`a_x. The double counting $E_x^{\rm SCAN,1c}$ uses the onsite τ of the local orbitals, is spin scaled, and by default contains exchange only (`TSCANCORR=T` adds SCAN correlation). Exchange-only spin scaling and the spin-polarized correlation of Chapter 2 apply as for SCAN.

Chapter 6

Self-consistency: the variational optimizer

6.1 The problem with density mixing

The standard SCF cycle alternates between solving the Kohn–Sham equations in a fixed potential and building a new input potential by mixing the input and output densities or potentials (Pulay/DIIS [24], Anderson [25], with Kerker preconditioning [26]). The energy along this path is a Harris–Foulkes-type estimate [27, 28]; it is not an upper bound and need not decrease. Mixing can stagnate, oscillate, or converge to an excited state, and nothing in the iteration signals it. VESPAW offers density mixing (`MIXER='PULAY'` or `'ANDERSON'`) but uses a variational optimizer by default.

6.2 Free energy as a function of the Hamiltonian

We minimize the Mermin free energy, Eq. (2.6), in the PAW form. The parameters of the Hamiltonian are collected in a vector

$$\boldsymbol{\theta} = \big(\tilde{v}(\boldsymbol{r}),\ v_\tau(\boldsymbol{r}),\ \{\Delta H_{R,ij}\}\big). \tag{6.1}$$

For given $\boldsymbol{\theta}$ we determine the eigenstates $\psi_n(\boldsymbol{\theta})$ of $\tilde{H}[\boldsymbol{\theta}]$ and the Fermi occupations $f_n(\boldsymbol{\theta})$ at fixed N, and define

$$\mathcal{A}(\boldsymbol{\theta}) = A\big[\{\psi_n(\boldsymbol{\theta})\}, \{f_n(\boldsymbol{\theta})\}\big]. \tag{6.2}$$

Because $\psi_n(\boldsymbol{\theta})$ and $f_n(\boldsymbol{\theta})$ form an admissible ensemble, $\mathcal{A}(\boldsymbol{\theta})$ is an upper bound of the minimum of A for every $\boldsymbol{\theta}$. At self-consistency, where $\boldsymbol{\theta}$ equals the parameters $\boldsymbol{v}_{\mathrm{out}}[\boldsymbol{\theta}]$ built from the output state, the bound is attained. Following Gonze [29],

$$\frac{\partial \mathcal{A}}{\partial \boldsymbol{\theta}} = \chi(\boldsymbol{\theta})\,\big(\boldsymbol{v}_{\mathrm{out}}[\boldsymbol{\theta}] - \boldsymbol{\theta}\big), \tag{6.3}$$

with χ the static independent-particle response. Since χ is negative semidefinite, the residual $\boldsymbol{r} = \boldsymbol{v}_{\mathrm{out}} - \boldsymbol{\theta}$, and any positive-definite preconditioned version $K\boldsymbol{r}$, is a descent direction; χ never has to be built. The Kerker filter $K(G) = G^2/(G^2 + q_0^2)$ is applied to the plane-wave part of the residual, with q_0^2 set by `MIXGOSQ` (default 1.5 bohr^{-2}). The default is made for solids. For molecules and slabs with large vacuum it damps the long-wavelength components so strongly that the gains of a step fall below the noise of the iterative eigensolver, and the line search stalls; use $q_0^2 \approx 0.05$–0.1 there.

Algorithm 1 Hamiltonian-space variational SCF (Route 1).

$\boldsymbol{\theta}_0 \leftarrow \boldsymbol{v}_{\text{out}}[\psi_{\text{start}}]$; hold fixed for a short eigensolver warm-up
loop
  diagonalize $\tilde{H}[\boldsymbol{\theta}]$ (Davidson, RMM-DIIS or CG); Fermi occupations at fixed N
  evaluate $\mathcal{A}(\boldsymbol{\theta})$ at the output state; build $\boldsymbol{v}_{\text{out}}$
  $\boldsymbol{r} \leftarrow \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$; project out $\boldsymbol{s}$
  if $\mathcal{A}(\boldsymbol{\theta}) \le \mathcal{A}_{\text{best}} + \epsilon$ **then**
    accept; add $(\boldsymbol{\theta}, K\boldsymbol{r})$ to the history; stop if converged
    $\boldsymbol{\theta} \leftarrow$ Anderson proposal
  else
    $\boldsymbol{\theta} \leftarrow \frac{1}{2}(\boldsymbol{\theta} + \boldsymbol{\theta}_{\text{best}})$ ▷ re-anchor after 5 rejections
  end if
  re-gauge $\boldsymbol{\theta}$ along $\boldsymbol{s}$ to the mean grid potential of $\boldsymbol{v}_{\text{out}}$
end loop

6.3 The gauge direction

A uniform shift $\tilde{H} \to \tilde{H} + c\hat{O}$ shifts all eigenvalues and the Fermi level by c and leaves orbitals and occupations unchanged. In parameter space it is the direction $\boldsymbol{s} = (1, 0, \{\Delta O_{R,ij}\})$, and $\mathcal{A}(\boldsymbol{\theta} + c\boldsymbol{s}) = \mathcal{A}(\boldsymbol{\theta})$ exactly. The two halves of $\boldsymbol{s}$ are updated by different rules (Kerker on the grid, damped steps in ΔH). If $\boldsymbol{s}$ is not treated explicitly, they drift apart and act as a spurious potential step between spheres and interstitial (Si: convergence to an excited state 28 mH above the ground state). VESPAW therefore

1. projects $\boldsymbol{s}$ out of every residual, and
2. after each step moves $\boldsymbol{\theta}$ along $\boldsymbol{s}$ so that its mean grid potential equals that of the output state.

The second step does not change any energy, but without it $\boldsymbol{\theta}$ keeps its starting gauge while the true potential moves: eigenvalues and Fermi level are then offset (0.49 eV in Si), and with them the Lagrange multipliers in forces and stress (Chapter 8).

6.4 Line search and acceptance

Steps are proposed by an Anderson extrapolation built from *accepted* points only. A trial $\boldsymbol{\theta}'$ is accepted if $\mathcal{A}(\boldsymbol{\theta}') \le \mathcal{A}_{\text{best}} + \epsilon$; otherwise the step is halved towards the best point. After five consecutive rejections the best point is re-evaluated ("re-anchoring"), because its energy may stem from an eigensolution that was not fully converged.

Convergence is reached when the mean-square grid residual is below `MIXTOL` (and the one-center residual below its square root), or when two consecutive accepted energies differ by less than `EDIFF`. Route 1 is selected by `!PSIDYN MIXER='VARIATIONAL'` together with `OPTIMIZER='DAV'`, `'RMM'` or `'CG'`; it is the default mixer.

Hybrid operators. The onsite hybrid operator of Chapter 5 depends on the orbitals through D, not only on the density. It is taken from the current state rather than line-searched; including its matrix elements in $\boldsymbol{\theta}$ left a residual that did not decay.

6.5 Route 2: direct joint minimization

As an alternative (`OPTIMIZER='EDFT'`), A is minimized directly with respect to the orbitals and an auxiliary Hamiltonian H^{aux} whose eigenvalues define the occupations, in the spirit of

ensemble DFT [30, 31]. The orbital direction is the Teter-preconditioned [32] projected residual combined by Polak–Ribière conjugate gradients; the direction for H^{aux} is $H^{\mathrm{phys}} - H^{\mathrm{aux}}$. Trial points are built by Löwdin orthonormalization and rotated into the eigenbasis of H^{aux}; step lengths come from a quadratic model, with the same acceptance and re-anchoring logic as Route 1. The orbital direction is computed from a Kerker-preconditioned Hamiltonian. With this preconditioning Route 2 currently stops about 5×10^{-6} H above the minimum on Si; without it (`VESPAW_EDFTVAR_NOKERKER=1`) it reaches the minimum.

6.6 The consistency test

The variational optimizer minimizes the energy. Damped Car–Parrinello dynamics (Chapter 7) and density mixing converge to a self-consistent state. The two coincide only if the Hamiltonian is the derivative of the energy. Agreement between them, to 10^{-6} H or better, is therefore a sharp test of any new functional or implementation: a missing potential term shows up as the variational optimizer finding energies *below* the self-consistent state without converging. The regression suite (Chapter 9) applies this test to every case.

Chapter 7

Eigensolvers and dynamics

For Route 1 the eigenstates of $\tilde{H}[\boldsymbol{\theta}]$ must be found at every iteration. They are warm-started from the previous iteration and converged iteratively. All solvers work with the generalized problem $\tilde{H}\psi = \epsilon\hat{O}\psi$ and apply $\tilde{H}$ and $\hat{O}$ in blocks of bands.

7.1 Block Davidson

`OPTIMIZER='DAV'`. Each band is extended by a preconditioned correction vector $(\tilde{H} - \epsilon\hat{O})\psi$; the new vectors are projected out of *all* current bands (twice), vectors lost to cancellation are dropped, and the extended basis is canonically orthogonalized with an eigenvalue cutoff of 10^{-10}. A Rayleigh–Ritz step in this subspace gives the new bands. Converged bands are locked. The implementation keeps seven wave-function-sized arrays. Davidson is robust from any start and is the reference solver.

Pitfall. An earlier Davidson orthogonalized only against converged bands and stabilized the near-singular overlap with a shift; the returned orbitals were not orthonormal, which produced energies below the minimum and split degenerate states.

7.2 RMM-DIIS

`OPTIMIZER='RMM'`. Residual minimization with direct inversion in the iterative subspace [33, 34] minimizes the norm of the residual of each band separately, without orthogonalization against other bands, and is therefore cheap and parallel. It converges to the eigenvector nearest the starting guess. Started from random orbitals it converges to wrong states, so the first `RMMDELAY` iterations (default 8) use Davidson. Bands are processed in blocks of 32 (`NRMMX` inner sweeps). RMM-DIIS is the fastest option for large systems (Si_{64}: 247 s vs 386 s for Davidson).

Pitfall. With the onsite hybrid, RMM-DIIS reaches the correct energy but can miss the lowest empty band (MgO PBE-SL0r gap 6.54 instead of 4.23 eV). Take band gaps of hybrid calculations from Davidson.

7.3 Conjugate gradients

`OPTIMIZER='CG'`. Band-by-band preconditioned conjugate-gradient minimization of the Rayleigh quotient with orthogonalization to the lower bands, followed by a subspace rotation. Reliable, slower than the block methods for many bands.

7.4 Car–Parrinello dynamics with friction

The original CP-PAW optimizer (`OPTIMIZER='MD'`, the default when no eigensolver is chosen) treats the wave-function coefficients as classical degrees of freedom with a fictitious mass [35],

$$m_\psi \ddot{\tilde{\psi}}_n = -\tilde{H}\tilde{\psi}_n + \sum_m \hat{O}\tilde{\psi}_m \Lambda_{mn} - \gamma\, m_\psi \dot{\tilde{\psi}}_n, \tag{7.1}$$

with Lagrange multipliers Λ that keep the orbitals orthonormal and a friction γ (`!PSIDYN FRIC`). With friction the dynamics relaxes into the ground state. The "autopilot" (`!PSIDYN!AUTO`) lowers the friction when the energy decreases and raises it when it increases. Damped dynamics is slow but very robust, converges to the self-consistent state for any functional, and serves as the reference in the consistency test. `SAFEORTHO=T` orthogonalizes such that the energy is conserved; it should not be used with variable occupations.

Chapter 8

Forces, stress and molecular dynamics

8.1 Forces

The force on atom R is the total derivative of the energy with respect to its position. At a stationary point of the energy the orbital response drops out (Hellmann–Feynman [36]). In PAW the projectors, the compensation charges and the one-center terms move with the atom, and the orthonormality constraint depends on the positions through $\hat{O}$:

$$\boldsymbol{F}_R = -\frac{\partial E}{\partial \boldsymbol{R}}\Big|_{\text{explicit}} - \sum_{nm} \Lambda_{mn} \langle \tilde{\psi}_n | \frac{\partial \hat{O}}{\partial \boldsymbol{R}} | \tilde{\psi}_m \rangle \times (\text{occupation weights}), \tag{8.1}$$

where the first term contains the derivatives of ΔH, the projectors and the compensation charges. In Car–Parrinello dynamics Λ comes from the orthonormality constraint. For the eigensolvers the orbitals are Ritz vectors, and $\Lambda = \mathrm{diag}(\epsilon_n)$.

Pitfalls. Before September 2026 the eigensolvers did not set Λ at all, so the constraint term was missing (0.5% error in the Si force), and with the variational mixer the eigenvalues carried the gauge offset of the previous chapter. Both are now fixed, and forces agree with finite differences of the energy to 10^{-4} relative (regression case `si_forces`).

8.2 Stress

The stress tensor is the derivative of the energy with respect to a homogeneous strain ϵ,

$$\sigma_{\alpha\beta} = -\frac{1}{V}\frac{\partial E}{\partial \epsilon_{\alpha\beta}}, \tag{8.2}$$

following Nielsen and Martin [37]; positive values mean that the cell wants to expand, and the pressure is $\mathrm{tr}\,\sigma/3$. It contains kinetic, Hartree, exchange-correlation, projector and compensation-charge terms, and the same constraint term as the forces.

The analytic stress is taken at a fixed set of plane waves (the basis strains with the cell). A finite difference at fixed cutoff changes the basis and contains the *Pulay stress*, which vanishes as the cutoff grows (Si: 2% at 25 Ry, 0.5% at 50 Ry). Lattice constants from energy-volume curves need converged cutoffs for the same reason.

Output. `!CONTROL!GENERIC FORCES=T STRESS=T` evaluates forces and stress also for fixed atoms and a fixed cell and prints them at full precision at the end of the run (`FORCE`, `STRESS`, `PRESSURE[GPA]` lines).

8.3 Structure relaxation and molecular dynamics

Atomic motion is controlled by `!CONTROL!RDYN`. With friction (`FRIC`, `!RDYN!AUTO`) the atoms relax into a minimum; without friction they follow Newton's equations. The cell shape can be relaxed or propagated with `!CONTROL!CELL MOVE=T` [37].

Car–Parrinello vs Born–Oppenheimer MD. In Car–Parrinello MD the wave functions are propagated together with the atoms and stay close to the ground state if their fictitious mass is small enough (adiabaticity [38]). The time step must resolve the fastest wave-function motion. In Born–Oppenheimer MD the electrons are converged at each step (with an eigensolver and Route 1); the time step is set by the ions, but each step costs a full SCF.

Thermostats. Temperature is controlled by Nosé–Hoover thermostats [39] on the atoms (`!RDYN!THERMOSTAT T[K]`) and, in Car–Parrinello MD, on the wave functions [40] (`!PSIDYN!THERMOSTAT`), which removes the heat that leaks from the ions into the fictitious electronic motion.

Chapter 9

Structure of the code

9.1 Source files

File	Content
`vespaw.f90`, `vespaw_driver.f90`	main program, time-step loop, reports
`vespaw_ioroutines.f90`	reading `.cntl/.strc` input
`vespaw_waves1.f90`, `vespaw_waves2.f90`	wave functions, density, $H\psi$, forces, total energy
`vespaw_cg.f90`	Davidson, RMM-DIIS, CG
`vespaw_mixer.f90`	variational, Pulay and Anderson mixers
`vespaw_edft.f90`	Route 2 (EDFT)
`vespaw_potential.f90`	plane-wave potential, electrostatics, XC on the grid
`vespaw_augmentation.f90`	one-center (PAW sphere) terms
`vespaw_setups.f90`	setup construction and tests
`vespaw_lmto.f90`	local orbitals, PBE0r, PBE-SL0r, SCAN0r
`vespaw_scantau.f90`, `vespaw_scannsph.f90`	SCAN with orbital τ, non-spherical one-center SCAN
`vespaw_dft.f90`	exchange-correlation functionals
`vespaw_atoms.f90`, `vespaw_cell.f90`	ionic and cell dynamics
`vespaw_occupations.f90`	occupations, Mermin

The code is organized in CP-PAW's object style: each module keeps its data in a Fortran module and is accessed through `OBJECT$SET`, `$GET` and action routines.

9.2 One SCF step

1. `PAW` (driver) calls `TIMESTEP`, which calls `WAVES$ETOT`.
2. `WAVES$ETOT`: projections $\langle\tilde{p}|\tilde{\psi}\rangle$, density and one-center density matrices; plane-wave potential (`WAVES_VOFRHO`); one-center potentials and ΔH (`AUGMENTATION`); hybrid terms (`LMTO`); forces and stress (`WAVES$FORCE`).
3. Variational mixer (`WAVES_VARMIXHAMILTONIAN`): the output Hamiltonian and the energy are handed to the line search, which returns $\boldsymbol{\theta}$ for the next diagonalization.
4. Eigensolver (`DAVIDSON$STATE_BY_BLOCK`, `RMM$STATE_BY_BLOCK`, `CG$STATE_BY_STATE`) or, for dynamics, $H\psi$ and propagation.
5. Reports (`PRINFO`), capture of forces and stress, and the switch of all dynamical variables to the next step.

9.3 Input

The `.cntl` and `.strc` files are read into linked lists (`LINKEDLIST$...`). Blocks start with `!NAME` and end with `!END`; a block whose name is changed (e.g. `!ORBPOT_OFF`) is ignored. `READ=F` inside a block also switches the block off. The routines `READIN_*` in `vespaw_ioroutines.f90` transfer the values to the modules.

9.4 Adding a functional

1. Implement energy density and potential in `vespaw_dft.f90` (or a module of its own), register a type name in `READIN_DFTTYPEFROMNAME`.
2. Provide the one-center evaluation (`vespaw_augmentation.f90`), with the full potential, Eq. (2.4).
3. Verify: energy vs finite difference of the potential along a density change; variational optimizer vs damped dynamics (Sec. 6.6); forces and stress vs finite differences.
4. Add a regression case.

9.5 Adding an optimizer

Optimizers are dispatched on `OPTIMIZERTYPE` in `WAVES$ETOT`. A new eigensolver must return orthonormal Ritz vectors and their Ritz values in `THIS%EIGVAL` (used for the Lagrange multipliers of the forces). It should be tested against Davidson on the whole regression suite.

9.6 Regression suite

`regression/run.py` runs all cases (`regression/*/case.json`) or selected ones (`run.py CASE:RUN`). A case has template inputs with `@PLACEHOLDER@` substitutions, several runs (optimizers, cold and seeded starts, multi-stage runs sharing a restart file), checks against reference values, agreement tests between runs, finite-difference tests of forces and stress, and a list of known issues. The cases cover PBE Si (all optimizers), hot Al (metal, Mermin), Si_8 at Γ, NiO AFM (PBE, PBE+U, PBE0r, PBE-SL0r, SCAN0r, TSCAN), MgO PBE-SL0r, TSCAN Si, spin–orbit Si, a polar slab with dipole correction, space-group symmetry, and forces and stress. Every change to the code should leave the suite passing.

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